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Stroop Task Experiment

PSY310 Lab In Psychology

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Introduction

The roles of attention and control of cognition are crucial to human information processing. Perhaps the strongest illustration of selective attention is the Stroop effect (Stroop, 1935). The Stroop task examines the impact that automatic avenues, i.e., the reading of words, have on more-controlled mechanisms, i.e., the naming of colors. Responses tend to be faster when the meaning of a word corresponds with the color the font of the word is written in (e.g. the word RED written in red ink). But in cases where the meaning and color of the question does not match (e.g., the word RED was written in blue ink), response time goes slower and errors are natural.

This effect reveals the automaticity of reading and how hard it is to inhibit the most dominant response in order to adopt another less automatic response. The Stroop Task has far reaching implications in the study of executive control, attention distribution, and processing velocity, which make it applicable in clinical and cognitive psychology.

Method

Participants

The research was conducted on 3 respondents aged, between 20-21 years all being students in PSY310 course in the Ahmedabad University. Informed consent was notified prior to the experiment and confidentiality was observed.

Materials

The experiment was run on a laptop with a 13.3-inch screen, 2560x1600 resolution and 60 Hz refresh rate by using PsychoPy v2025.1.1.

The stimuli- the letters Red and Blue were presented in both the fonts.

Two main criteria were selected:

- Congruence trials: Word meaning and letter color were the same (e.g., “RED” with red lettering).
- Incongruent trials: Word meaning and the color of the font were different (e.g., “RED” in blue letters).

Procedure

The participants sat comfortably and they were informed that they should identify the font color of each word as fast and accurately as possible by pressing the specified keys (e.g., to press the key R in case of red, B in case of blue). It is not encouraged to read the word and name the color itself.

In the beginning of each trial, the fixation cross appeared on the screen during 500 ms and then the stimulus was shown during 2000 ms or until response. The study was performed using 40 trials, half consisted of congruent and the other one of incongruent, in random order. There were Reaction Time (RT) and accuracy data saved on each trial.

Results

PARTICIPANT	CONGRUENT RT	INCONGRUENT RT	CONGRUENT ACCURACY (%)	INCONRUENT ACCURACY (%)	STROOP EFFECT
1	1.022508355	0.892851845	100	100	-0.12965651
2	0.49687299	0.66057277	95	85	0.16369978
3	0.864000035	1.45509499	100	95	0.591094955
	0.79446046	1.002839868	98.33333333	93.33333333	0.208379408

- Accuracy is higher in Congruent (i.e. 98.33%) than Incongruent (i.e. 93.33%).
- Stroop effect = 0.20837

Discussion

The findings confirm the hypothesis and prove the existence of a Stroop effect whereby the participants need a longer time in incongruent trials as compared to congruent trials in naming the color. This result concurs with the initial study by Stroop (1935) and other studies that demonstrated that automaticity in reading words disrupts the naming of colors and more cognitive control is needed to overcome conflict. These findings indicate the importance of selective attention and executive functioning in everyday life, where one has to block out the irrelevant but salient information.

The limitations are possible because of a relatively small sample and the application of only one language (English) that can influence the generalization. Future study might address the question of whether bilinguals exhibit a reduced degree of Stroop interference and whether Stroop performance differs when cognitive load is applied